

EXPRESSIONS, VARIABLES, OPERATORS

DEMO: RUN JAVASCRIPT PROGRAM "HELLO WORLD" IN BROWSER

```
alert('Hello world');
```

DEMO: RUN JAVASCRIPT PROGRAM "HELLO WORLD" IN NODE

```
process.stdout.write('Hello world');
```

DEFINION

*JavaScript language **doesn't exist by itself**. It's a scripting language that **must be hosted elsewhere**.*

DEFINITION

JavaScript language - is something like an abstraction

RunTime - is a program which implements the
abstraction mentioned above

Host - is an another program that has RunTime inside

HOST AND RUNTIME

- **RunTime**: V8, SpiderMonkey, JavaScript Core, etc...
- **Host**: Browser, Node, d8, etc...

JAVASCRIPT LANGUAGE OWNS:

```
AggregateError;  
Array;  
ArrayBuffer;  
Atomics;  
BigInt;  
BigInt64Array;  
BigUint64Array;  
Boolean;  
DateView;  
Date;  
decodeURI;  
decodeURIComponent;  
encodeURI;  
encodeURIComponent;  
Error;
```

DON'T EXIST IN JAVASCRIPT LANGUAGE

```
DOM;  
Events;  
EventLoop;  
setTimeout, setInterval;  
fs;  
canvas  
console.log;  
alert;  
Garbage Collector;  
...
```

DEFINITION

Host could live without scripting language JavaScript (i.e Browser without JS could still display web pages)

JAVASCRIPT

THIS **IS NOT** JAVASCRIPT

```
alert('Hello world');
```

THIS **IS** JAVASCRIPT

```
'Hello world';
```

*Be able to spot the difference between
JavaScript language features and host
features*

EXPRESSIONS

```
> 5  
< 5
```

```
> 7  
< 7
```

```
> 5 + 7  
< 12
```

EXPRESSIONS

```
> var sum = 5 + 7;  
< undefined
```

```
> var sum = 5 + 7;  
< sum;
```

EXPRESSIONS

```
> var a = 5;  
   var b = 7;  
   a + b;  
< 12
```

```
> var a = 5;  
   var b = 7;  
   var sum = 5 + 7;  
< undefined
```

EXPRESSIONS

```
> (5 + 7)  
< 12
```

```
> (5 + 7) ;  
< 12
```

EXPRESSIONS

In JavaScript what you see is something that is being computed, it is not what it seems to be

- a number is not a number, it's an expression to be computed
- () - is an expression
- ; is an expression

DEFINITION

Variable - is a container to store a link and a data,
accessible by this link

All JavaScript variables must be **identified** with
unique names

These unique names are called **identifiers**

IDENTIFIERS

Identifiers can be short names (like x and y) or more descriptive names (age, sum, totalVolume).

The general rules for constructing names for variables (unique identifiers) are:

- Names can contain letters, digits, underscores, and dollar signs
- Names must begin with a letter
- Names can also begin with \$ and _
- Names are case sensitive (y and Y are different variables)
- Reserved words (like JavaScript keywords) cannot be used as names



VARIABLE DECLARATION

JavaScript Variables can be declared in 3 ways:

- Using **var**
- Using **let**
- Using **const**

VARIABLE DECLARATION: **VAR**

```
var x;  
var y;  
var z;
```

VARIABLE DECLARATION: LET

```
let x;  
let y;  
let z;
```

VARIABLE DECLARATION: **CONST**

```
const x; // error  
const y; // error  
const z; // error
```

VARIABLE DECLARATION AND INITIALIZATION: **VAR**

```
var x = 5;  
var y = 6;  
var z = x + y;
```

VARIABLE DECLARATION AND INITIALIZATION: **LET**

```
let x = 5;  
let y = 6;  
let z = x + y;
```

VARIABLE DECLARATION AND INITIALIZATION: **CONST**

```
const x = 5;  
const y = 6;  
const z = x + y;
```

DECLARATION AND INITIALIZATION

```
// let x - is a part called declaration  
// = 24 - is a part called initializer  
let x = 24;
```

In var and let declarations initializer part is options. In
const it's required;

```
let x;  
var y;  
const z = x + y;
```

ASSIGNMENT (LINKING) OPERATOR =

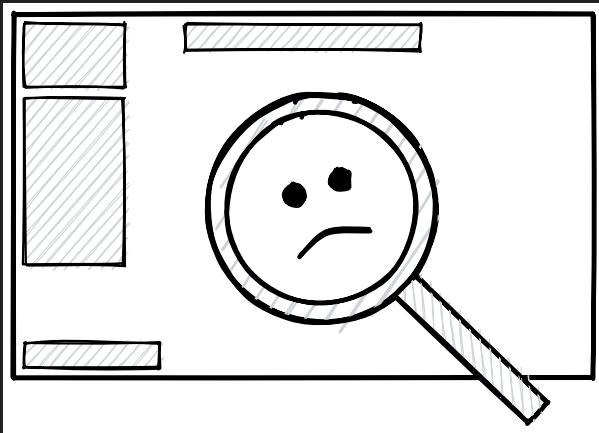
```
let x = 24;
```

```
let x = 5;  
let y = 7;  
const z = x + y;
```

WHY ASSIGNMENT IS LINKING?



EXCALIDRAW+



Scene Not Found

ONE STATEMENT, MANY VARIABLES

You can declare many variables in one statement.

Start the statement with `let` and separate the variables by comma:

```
let two = 2,  
    model = 6542,  
    price = 200;
```

VALUE = UNDEFINED

A variable declared without a value will have the value **undefined**

```
> let price;  
> price;  
< undefined
```

```
> const price = 500;  
> price;  
< 500
```

RE-DECLARING JAVASCRIPT VARIABLES

You cannot re-declare a variable declared with `let` or `const`

```
> let price = 500;  
> let price;  
> price;  
< Uncaught SyntaxError: Identifier 'price' has already been de
```

If you re-declare a variable declared with `var`, it will not lose its value.

```
> var price = 250;  
> var price;  
< 250
```

USED MATERIALS

<https://www.youtube.com/watch?v=1F-8pn30bOI>

https://www.youtube.com/watch?v=U5BN_lFE5d8&list=PL3ziSA8uO7Kn

https://www.w3schools.com/js/js_variables.asp

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